

Polynomial End Behavior, Zeros & Multiplicity

ANSWER KEY



End behavior of polynomial functions

- 1 State the end behavior of $f(x) = -2x^5 + 3x^2 - 1$, using the leading term.
Leading term $-2x^5$: odd degree, negative leading coefficient. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$.
- 2 State the end behavior of $f(x) = 3x^4 - 5x^3 + 2$.
Leading term $3x^4$: even degree, positive leading coefficient. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$.

Zeros and their multiplicities

- 3 Find the zeros of $f(x) = (x - 2)^3(x + 1)^2$ and state the multiplicity and graphical behavior at each.
 $x = 2$ has multiplicity 3 (odd): the graph crosses the axis, flattening near $x = 2$. $x = -1$ has multiplicity 2 (even): the graph touches the axis and turns back (a local extremum), not crossing.
- 4 The graph shows $f(x) = (x + 2)(x - 1)^2$, illustrating a crossing zero at $x = -2$ and a touching zero at $x = 1$.
At $x = -2$ (multiplicity 1, odd): the graph crosses the x -axis. At $x = 1$ (multiplicity 2, even): the graph touches the axis and turns back without crossing, consistent with the flattened touch point shown.

The Rational Zero Theorem

- 5 List all possible rational zeros of $f(x) = 2x^3 - 3x^2 - 8x + 12$ using the Rational Zero Theorem, then find the actual zeros.
Possible rational zeros: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm \frac{1}{2}, \pm \frac{3}{2}$ (factors of 12 over factors of 2). Testing $x = 2$: $2(8) - 3(4) - 16 + 12 = 16 - 12 - 16 + 12 = 0$ ✓. Dividing:
 $2x^3 - 3x^2 - 8x + 12 = (x - 2)(2x^2 + x - 6) = (x - 2)(2x - 3)(x + 2)$. Zeros: $x = 2, \frac{3}{2}, -2$.

Constructing a polynomial from its zeros

- 6 Write a polynomial function of least degree with real coefficients, zeros at $x = -1$ (multiplicity 2) and $x = 3$ (multiplicity 1), and a leading coefficient of 1.
 $f(x) = (x + 1)^2(x - 3) = (x^2 + 2x + 1)(x - 3) = x^3 - x^2 - 5x - 3$.
- 7 A polynomial has zeros at $x = 0$, $x = 2$, and $x = -3$, all with multiplicity 1, and passes through $(1, 8)$. Find the polynomial in the form $f(x) = a \cdot x(x - 2)(x + 3)$.
 $f(1) = a(1)(1 - 2)(1 + 3) = a(1)(-1)(4) = -4a = 8$, so $a = -2$. $f(x) = -2x(x - 2)(x + 3)$.

Local extrema and turning points

- 8 A polynomial of degree n has at most how many turning points (local extrema)? State the rule and apply it to a degree-5 polynomial.

A degree- n polynomial has at most $n - 1$ turning points. A degree-5 polynomial has at most 4 turning points.

- 9 The graph of a polynomial shows exactly 3 turning points. What is the minimum possible degree of the polynomial?

Since turning points $\leq n - 1$, having 3 turning points requires $n - 1 \geq 3$, so $n \geq 4$. The minimum degree is 4.

Average rate of change and concavity

- 10 Find the average rate of change (AROC) of $f(x) = x^3 - 4x$ on $[1, 3]$.

$$\text{AROC} = \frac{f(3) - f(1)}{3 - 1} = \frac{(27 - 12) - (1 - 4)}{2} = \frac{15 - (-3)}{2} = \frac{18}{2} = 9.$$

- 11 For $f(x) = x^3 - 4x$, compare the AROC on $[0, 1]$ and $[1, 2]$, and use the comparison to determine whether f is concave up or concave down on this interval.

AROC on $[0, 1]$: $\frac{f(1) - f(0)}{1 - 0} = \frac{-3 - 0}{1} = -3$. AROC on $[1, 2]$: $\frac{f(2) - f(1)}{2 - 1} = \frac{0 - (-3)}{1} = 3$. Since the AROC is increasing from -3 to 3 , f is concave up on $[0, 2]$.

Sign charts and interval analysis

- 12 Build a sign chart for $f(x) = (x + 1)(x - 2)(x - 4)$ and state the intervals where $f(x) > 0$.

Zeros at $x = -1, 2, 4$ split the line into four intervals. Testing a point in each: $x < -1$ gives $f < 0$; $-1 < x < 2$ gives $f > 0$; $2 < x < 4$ gives $f < 0$; $x > 4$ gives $f > 0$. So $f(x) > 0$ on $(-1, 2) \cup (4, \infty)$.

- 13 Use a sign chart to solve the inequality $(x - 1)^2(x + 3) \leq 0$.

Zeros at $x = 1$ (multiplicity 2, sign does not change) and $x = -3$ (multiplicity 1, sign changes). Testing intervals: for $x < -3$, expression is negative; for $-3 < x < 1$, positive; for $x > 1$, positive. So the expression is ≤ 0 on $(-\infty, -3] \cup \{1\}$.

Visualizing end behavior families

- 14** The graph shows $f(x) = x^4 - 5x^2 + 4$, a degree-4 polynomial with four real zeros.
- a** State the end behavior of the graph as $x \rightarrow \pm \infty$. Even degree, positive leading coefficient: as $x \rightarrow \pm \infty$, $f(x) \rightarrow +\infty$, matching the upward arms shown on both sides.
 - b** Identify the four zeros shown and state whether each is a crossing or touching zero. Zeros at $x = -2, -1, 1, 2$, each multiplicity 1 (odd), so all four are crossing zeros.
 - c** How many turning points does the graph display, and is this consistent with the degree? The graph shows 3 turning points, which is exactly $n - 1 = 4 - 1 = 3$ for a degree-4 polynomial — consistent with the maximum allowed.
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Putting it together: a full analysis

- 15** For $f(x) = x^3 + 2x^2 - 5x - 6$: verify that $x = -1$ is a zero, factor completely, then describe the end behavior and the behavior at each zero.
- $f(-1) = -1 + 2 + 5 - 6 = 0$ ✓. Dividing by $(x + 1)$:
- $x^3 + 2x^2 - 5x - 6 = (x + 1)(x^2 + x - 6) = (x + 1)(x + 3)(x - 2)$. Zeros at $x = -1, -3, 2$, all multiplicity 1 (crossing). End behavior: odd degree, positive leading coefficient, so $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$.